

Nicholas Channg

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EDUCATION

Cornell University

Ithaca, NY

B.A. Computer Science, Minors in Artificial Intelligence, Data Science | GPA: 3.72/4.0 Expected Graduation, May 2027

- **Awards:** 4x Hackathon Winner, Y Combinator AI Startup School 2025 (1 of 2,500 selected)

EXPERIENCE

Google

Sunnyvale, CA

Incoming Software Engineer Intern, Google Cloud Knowledge Catalog

Summer 2026

Amazon

New York, NY

Software Engineer Intern, Amazon Ads

Sep 2025 – Dec 2025

- Built and owned asynchronous Java and Python backend services in a distributed AWS video ads processing pipeline handling 80K+ jobs weekly, designing fault-tolerant workflows across SQS, ECS, Lambda, DynamoDB, and S3
- Developed a production AI microservice that performs multimodal video analysis via AWS Bedrock VLM/LLM, generating optimized ad titles/descriptions and improving text-asset relevance for 98% of video ads
- Integrated a video assembly service into the pipeline by building a Lambda-triggered ECS execution path and S3/DynamoDB asset storage, enabling secure cross-account and cross-service access of final video assets

Tesla

Fremont, CA

Software Engineer Intern, Charging Team

May 2025 – Sep 2025

- Developed scalable REST API endpoints in the Tesla Mobile App backend with Node.js, Express.js, and TypeScript, orchestrating vehicle/charging microservice calls to power real-time user-visible data for 3.4M daily users
- Implemented a Redis caching layer to store Spark API responses for supercharger congestion/pricing data, reducing redundant upstream microservice calls and improving Tesla Mobile App response latency by 10%
- Developed an agentic AI QA assistant with Python that automated mobile app test-account setup (e.g. region switching, vehicle linking) via orchestrated microservice API calls, improving QA throughput by 25%

Flow

Remote

Software Engineer Intern

Jan 2024 – May 2024

- Developed REST APIs using Flask and Python to retrieve and process data from Flow's AI models, automating customer prospecting and increasing sales representative call volume by 300% (30 → 120 calls/day)
- Reduced application latency by 15% by optimizing PostgreSQL queries; replaced nested SELECTs with JOINs, added B-tree indexes on high-cardinality columns, and batched execution to minimize database calls
- Integrated backend services into CI/CD pipelines using GitHub Actions, ensuring automated testing and deployment

PROJECTS

ElectAI | Python, Scikit-learn, Flask, React, TypeScript, TailwindCSS, NumPy, Pandas

Jan 2025

- Engineered a full-stack ML web app to predict voter turnout using historical behavioral data patterns to model participation likelihood using a Flask REST API to retrieve data from the model and a React frontend
- Built a data preprocessing pipeline with Python, NumPy, and Pandas to clean 20 years of election data, enabling accurate regression modeling with Scikit-learn and achieving high accuracy (RMSE: 5.35, MAE: 4.22)

MathGPT | React, TypeScript, HTML, CSS, Microsoft Phi-3 API

July 2024

- Built an AI Math Tutor Chrome extension using Microsoft's Phi-3 LLM, React, and TypeScript that provides personalized math tutoring to 100+ students, featuring structured prompts and step-by-step AI solutions
- Reduced latency by 67% (30s → 10s) by dynamically adjusting user input via prompt engineering and implementing conditional API calls, eliminating redundant requests to the LLM and timeout issues

TECHNICAL SKILLS

Languages: Java, Python, C, C++, TypeScript, JavaScript, SQL

Frameworks: Flask, FastAPI, Node.js, Express.js, React, Next.js

Cloud/Infrastructure: AWS, Redis, PostgreSQL, Docker, CI/CD

Libraries/Tools: Scikit-learn, NumPy, Pandas, OpenAI API, TailwindCSS, Git, Postman